Lecture 4: Binomial Trees and Random Walks

1-Step Model: Basic Definition

We begin with the simplest discrete-time model for stock price evolution.

Let:

- S_0: the initial stock price.
- S_1: the stock price after one time step.

The 收益 （return） after one time step is defined as:

$$\text{Return} = \frac{S_1}{S_0}.$$

也就是相对收益率（Gross return），gross return 是含本金的回报；如果要减去本金，就是 net return。

We can rewrite S_1 in terms of the return factor:

$$S_1 = \left(\frac{S_1}{S_0}\right) S_0.$$

未来价格 = (收益倍率) × (当前价格)
未来价格可以用当前价格乘以一个收益倍率来表达。

Interpretation:

- If $\frac{S_1}{S_0} > 1$, the stock has a **positive return**.
- If $\frac{S_1}{S_0} < 1$, the stock has a **negative return**.

The Binomial Model Setup

In the one-step binomial model, the stock price can either go up or down after one time step:

$$\frac{S_1}{S_0} = \begin{cases} 1+u & \text{with probability } p, \\ 1+d & \text{with probability } 1-p, \end{cases}$$

where:

- $u > 0$: 上涨幅度 up factor (stock price increases by u).
- $d < 0$: 下跌幅度 down factor (stock price decreases by $|d|$).

Therefore, S_1 can be written as:

$$S_1 = \begin{cases} (1+u)S_0 & \text{with probability } p, \\ (1+d)S_0 & \text{with probability } 1-p. \end{cases}$$

Define random variable K:

$$K = \frac{S_1}{S_0} = \begin{cases} 1+u & \text{with probability } p, \\ 1+d & \text{with probability } 1-p. \end{cases}$$

Sometimes, for simplicity, we redefine:

$$K = \begin{cases} u & \text{with probability } p, \\ d & \text{with probability } 1-p. \end{cases}$$

where u and d directly represent proportional changes.

Expected Return and Variance

对于任意离散型随机变量 X，其期望（均值）定义为：

$$\mathbb{E}[X] = \sum (\text{取值}) \times (\text{对应概率})$$

可以计算未来收益倍率K的期望, the expected value of K is:

$$\mathbb{E}[K] = pu + (1-p)d.$$

对于任意随机变量 K，方差定义为：

$$\text{Var}(K) = \mathbb{E}[K^2] - (\mathbb{E}[K])^2$$

计算 $\mathbb{E}[K^2]$：

$$\mathbb{E}[K^2] = p \cdot u^2 + (1-p) \cdot d^2$$

所以：

$$\text{Var}(K) = \left(pu^2 + (1-p)d^2\right) - (pu + (1-p)d)^2$$

The variance of K is:

$$\text{Var}(K) = pu^2 + (1-p)d^2 - (pu + (1-p)d)^2 .$$

方差衡量的是风险或波动性。

1-step 二项模型：未来价格只有两种可能（上涨或下跌）。
Binomial Tree 的核心思想：用简单的上下跳跃模型来模拟资产价格的可能路径；未来收益是随机变量，可以计算期望和方差。

Special Case: Symmetric Binomial Model ($p = \frac{1}{2}$)

Now consider the special case where the up and down movements are equally likely, i.e. $p = \frac{1}{2}$.

Then:

$$K = \begin{cases} u & \text{with probability } \frac{1}{2}, \\ d & \text{with probability } \frac{1}{2}. \end{cases}$$

现在假设上涨和下跌的概率各为一半,这是等概率二项模型（公平市场假设）的一个特例。
The expected value becomes:

$$\mathbb{E}[K] = \mu = \frac{u+d}{2}.$$

$$\mathrm{Var}(K) = \mathbb{E}[K^2] - (\mathbb{E}[K])^2$$

其中:

- $\mathbb{E}[K^2] = \dfrac{u^2+d^2}{2}$
- $(\mathbb{E}[K])^2 = \left(\dfrac{u+d}{2}\right)^2$

The variance is computed as:

$$\mathrm{Var}(K) = \frac{u^2+d^2}{2} - \left(\frac{u+d}{2}\right)^2.$$

Expanding and simplifying:

$$4\,\mathrm{Var}(K) = 2(u^2+d^2) - (u+d)^2 = u^2 - 2ud + d^2 = (u-d)^2,$$

so:

$$\mathrm{Var}(K) = \left(\frac{u-d}{2}\right)^2.$$

方差只和上下跳跃的距离有关。

标准差 Standard Deviation and Mean Representation

We define:

$$\sigma = \sqrt{\mathrm{Var}(K)} = \frac{u-d}{2},$$

上下两跳的距离的一半，就是标准差
and

$$\mu = \frac{u+d}{2}.$$

$$u = \mu + \sigma, \quad d = \mu - \sigma$$

Thus, we can rewrite K as:

$$K = \begin{cases} \mu + \sigma & \text{with probability } \frac{1}{2}, \\ \mu - \sigma & \text{with probability } \frac{1}{2}. \end{cases}$$

This representation centers the random variable K around its mean μ with symmetric deviation σ.

也就是把每次的跳跃用“均值±标准差”的形式来描述，这个形式是:

1. 更加对称;
2. 为后续多步累加时使用中心极限定理（CLT）做准备;
3. 也是很多金融模型（如 Cox-Ross-Rubinstein 二项模型）中使用的标准表达。

Binomial Tree Diagram

The one-step binomial model can be visualized as:

$$\begin{array}{ccc} & S_0 & \\ \swarrow & & \searrow \\ S_0(1+u) & & S_0(1+d) \end{array}$$

- Probability p leads to $S_0(1+u)$.
- Probability $1-p$ leads to $S_0(1+d)$.

This tree structure will later be extended to multi-step models.

二步二叉树模型 2-Step Binomial Tree Model

We now extend the one-step binomial tree to a two-step model. At each step, the stock price can either move up by factor u or down by factor d. The random return factor K at each step is:

$$K = \begin{cases} u & \text{with probability } p, \\ d & \text{with probability } 1-p. \end{cases}$$

Let S_0, S_1, and S_2 be the stock prices at times 0, 1, and 2, respectively.

总收益倍率 Stock Price After Two Steps

The total return over two steps is given by:

$$\frac{S_2}{S_0} = \frac{S_2}{S_1} \cdot \frac{S_1}{S_0}.$$

由于假设每一步独立（Assume independent），因此两步的收益相乘即可。

第一步	第二步	最终收益倍率
涨	涨	$(1+u)(1+u) = (1+u)^2$
涨	跌	$(1+u)(1+d)$
跌	涨	$(1+d)(1+u)$
跌	跌	$(1+d)(1+d) = (1+d)^2$

其中**涨跌次序不影响中间两种路径的结果**，因为乘法可交换。

Assuming that returns in each step are independent, the possible outcomes for $\frac{S_2}{S_0}$ are:

$$\frac{S_2}{S_0} = \begin{cases} (1+u)^2 & \text{with probability } p^2, \\ (1+u)(1+d) & \text{with probability } p(1-p) + (1-p)p = 2p(1-p), \\ (1+d)^2 & \text{with probability } (1-p)^2. \end{cases}$$

Binomial Tree Representation

The corresponding binomial tree can be illustrated as:

Note: Since $(1+u)(1+d) = (1+d)(1+u)$, the middle two branches are combined.

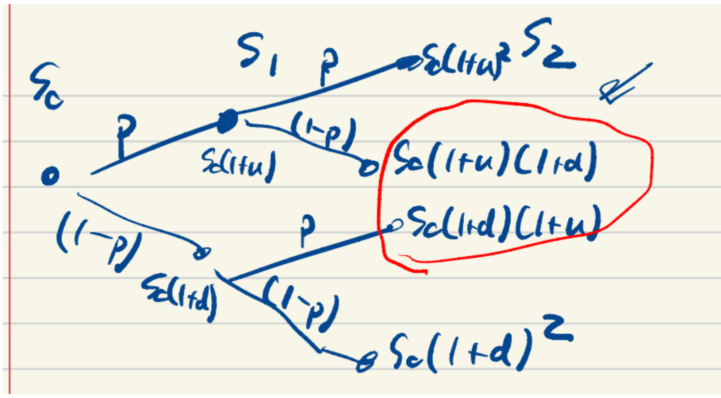

Abbildung 1: 二步二叉树示意图

计算期望值 Expected Value of $\frac{S_2}{S_0}$

We compute the expected value:

$$\mathbb{E}\left[\frac{S_2}{S_0}\right] = p^2(1+u)^2 + 2p(1-p)(1+u)(1+d) + (1-p)^2(1+d)^2.$$

We observe that this expression can be rewritten as:

$$\mathbb{E}\left[\frac{S_2}{S_0}\right] = \left(p(1+u) + (1-p)(1+d)\right)^2.$$

两步的期望收益倍率其实等于单步期望收益倍率的平方。
This result shows that the expected value of $\frac{S_2}{S_0}$ equals the square of the expected value of $\frac{S_1}{S_0}$. This is consistent with the fact that each period is independent and identically distributed.
多步二叉树模型的核心逻辑就是：每一步的收益是独立的，累积收益倍率就是各步倍率的乘积，而整个模型最终会逼近正态分布（中心极限定理）。

N-Step Binomial Tree Model

We now generalize to N time steps.

At each time step, the stock price can move up by factor u or down by factor d, independently of previous steps.

Let $S_0, S_1, \ldots, S_N$ denote the stock prices at each time step.

The cumulative return over N steps is:

$$\frac{S_N}{S_0} = \frac{S_1}{S_0} \cdot \frac{S_2}{S_1} \cdot \cdots \cdot \frac{S_N}{S_{N-1}}.$$

整个 N 步后的总收益倍率为：

$$\frac{S_N}{S_0} = \prod_{i=1}^{N} \frac{S_i}{S_{i-1}}$$

Since each ratio $\frac{S_{i+1}}{S_i}$ is independently equal to either $(1+u)$ or $(1+d)$, the possible outcomes for $\frac{S_N}{S_0}$ are:

$$\frac{S_N}{S_0} = (1+u)^i(1+d)^{N-i},$$

where i is the number of up moves.

- 你有 N 步。
- 每一步：
  - 可能上涨（成功）—— 概率 p
  - 可能下跌（失败）—— 概率 $1-p$
- 每次试验彼此独立。

你现在想求：

在 N 步中，恰好有 i 次上涨，$N-i$ 次下跌的概率是多少?

如果有一条路径里有 i 次上涨，$N-i$ 次下跌，那么这条路径的概率就是：

$$p^i \cdot (1-p)^{N-i}$$

因为：

- 每次上涨的概率是 p，连续乘 i 次；
- 每次下跌的概率是 $1-p$，连续乘 $N-i$ 次；
- 每一步独立，乘法法则成立。

Q：有多少条这样的路径？ 这个问题等价于：

- 在 N 个时间段中，**从中挑出 i 个时刻来上涨，其余下跌**；
- 这正好是组合数问题，答案是：
$$\binom{N}{i}$$

也就是：

$$\binom{N}{i} = \frac{N!}{i!(N-i)!}$$

The probability of each outcome follows the binomial distribution，任意一条涨 i 次、跌 $N-i$ 次的路径，概率都是 $p^i(1-p)^{N-i}$，而这样的路径有 $\binom{N}{i}$ 条，所以：

$$P\left(\frac{S_N}{S_0} = (1+u)^i(1+d)^{N-i}\right) = \binom{N}{i} p^i(1-p)^{N-i}.$$

Connection to 布朗运动 （Brownian Motion）

- 我们希望理解股票价格在连续时间上的分布行为；
- 离散模型是连续模型的近似。

基本思路：

- 把 N-step binomial tree 看作连续模型的近似；
- 每一步的 up 和 down 都取等概率 $p = \frac{1}{2}$（简化推导，核心思想不变）；
- 随着 $N \to \infty$，最终会导出连续模型。

Our ultimate goal is to understand how Brownian motion arises as a limit of the binomial model when N becomes large.

We are interested in studying the distribution of the stock price over time $[0, t]$.

We assume that the stock path distribution is the limit of discrete N-step binomial models, where for each N, we have up and down probabilities equal to $\frac{1}{2}$:

$$p = \frac{1}{2}.$$

Scaling Time and Step Size

We fix large $N \gg 0$ and define the time increment:

$$\Delta t = \tau = \frac{t}{N}.$$

- 我们把二项树模型看成把时间 $[0, t]$ 均匀切成 N 段；
- 每段长度是：
$$\Delta t = \frac{t}{N}$$
- 每一步价格涨跌的变化，就是在这么短的 Δt 里发生的涨或跌；
- 当 N 越来越大，Δt 越来越小；
- 最后当 $N \to \infty$ 时，就是连续模型。

Thus, each binomial step corresponds to a small time increment τ.

定义每步的收益倍率, at each step, the return factor K_N is defined as:

$$K_N = \begin{cases} u_N & \text{with probability } \frac{1}{2}, \\ d_N & \text{with probability } \frac{1}{2}. \end{cases}$$

Since the returns accumulate multiplicatively, and products naturally appear in expectations and variances, it is convenient to study the logarithm of the return(因为最终是乘积模型，为了简化分析，我们考虑对数收益):

$$\ln\left(\frac{S_t}{S_0}\right).$$

We define 每一步的对数收益:

$$k_N = \begin{cases} \ln(1+u_N) & \text{with probability } \frac{1}{2}, \\ \ln(1+d_N) & \text{with probability } \frac{1}{2}. \end{cases}$$

Log-Return Representation

We have:

$$\frac{S_t}{S_0} = \prod_{j=1}^{N} \frac{S_{j\tau}}{S_{(j-1)\tau}},$$

which gives 总的对数收益为:

$$\ln\left(\frac{S_t}{S_0}\right) = \sum_{j=1}^{N} \ln\left(\frac{S_{j\tau}}{S_{(j-1)\tau}}\right) = \sum_{j=1}^{N} k_N(j).$$

由于每步独立，所以总收益为 i.i.d. 随机变量的和。

Mean and Variance

The mean is:

$$\mu = \mathbb{E}\left[\sum_{j=1}^{N} k_N(j)\right] = N \cdot \mathbb{E}[k_N],$$

thus,

$$\mathbb{E}[k_N] = \frac{\mu}{N}.$$

The variance of $\ln\left(\frac{S_t}{S_0}\right)$ is:

$$\sigma^2 = \mathrm{Var}\left(\sum_{j=1}^{N} k_N(j)\right).$$

Since the terms $k_N(j)$ are independent and identically distributed (i.i.d.), we have:

$$\sigma^2 = N \cdot \mathrm{Var}(k_N).$$

Therefore:

$$\mathrm{Var}(k_N) = \frac{\sigma^2}{N}.$$

这里σ是连续极限中的总波动率。

Specification of k_N

当你把二项树模型的步数 N 越分越细（步长趋近于 0，步数趋近于无穷），你需要特别小心每步的上涨幅度和波动幅度怎么缩放，否则模型就会发散坍塌（爆炸或坍塌）。

核心缩放公式

每一步的小跳跃（对数收益）被设定为:

$$k_N = \begin{cases} \frac{\mu}{N} + \frac{\sigma}{\sqrt{N}} & \text{概率 } \frac{1}{2} \\ \frac{\mu}{N} - \frac{\sigma}{\sqrt{N}} & \text{概率 } \frac{1}{2} \end{cases}$$

也就是:

$$k_N = \frac{\mu}{N} \pm \frac{\sigma}{\sqrt{N}}$$

We specify that:

$$k_N = \begin{cases} \ln(1+u_N) = \frac{\mu}{N} + \frac{\sigma}{\sqrt{N}} & \text{with probability } \frac{1}{2}, \\ \ln(1+d_N) = \frac{\mu}{N} - \frac{\sigma}{\sqrt{N}} & \text{with probability } \frac{1}{2}. \end{cases}$$

This completes the binomial approximation before passing to the continuous-time limit.

Toward Brownian Motion: Central Limit Theorem Limit

We now introduce two auxiliary random variables that will help us transition to continuous-time Brownian motion.

Introducing ξ_N

Define a new random variable ξ_N 标准化的小跳跃:

$$\xi_N = \begin{cases} \frac{1}{\sqrt{N}} & \text{with probability } \frac{1}{2}, \\ -\frac{1}{\sqrt{N}} & \text{with probability } \frac{1}{2}. \end{cases}$$

Note that ξ_N is symmetric, mean zero, and variance $\frac{1}{N}$.

- 它是均值为 0，方差为 $\frac{1}{N}$ 的对称分布;
- 你可以把它看成是标准二项模型的标准化步长。

定义总跳跃 Sum of ξ_N: W_N

We now define:

$$W_N = \xi_N(1) + \xi_N(2) + \cdots + \xi_N(N),$$

where $\{\xi_N(i)\}_{i=1}^N$ are i.i.d. copies of ξ_N. Therefore, W_N is the sum of N i.i.d. random variables.

- 也就是 N 次独立 i.i.d. 的 ξ_N 累积和;
- 其实这是标准随机游走（random walk）的 rescaled 版本。

Expressing k_N in terms of ξ_N

Recall that earlier we approximated:

$$k_N = \frac{\mu}{N} + \sigma \cdot \xi_N.$$

This form clearly separates the drift term $\frac{\mu}{N}$ and the random fluctuation term $\sigma\xi_N$.
你把原来每一步的 log-return ξ_N 写成:

$$k_N = \frac{\mu}{N} + \sigma \cdot \xi_N$$

- 这里的 $\frac{\mu}{N}$ 是每一步的确定性漂移 (drift);
- $\sigma \cdot \xi_N$ 是波动性 (volatility) 部分;
- Discrete version of Geometric Brownian Motion。

Rewriting $\ln(S_t/S_0)$

The log-return can be expressed as:

$$\ln\left(\frac{S_t}{S_0}\right) = \sum_{i=1}^{N} k_N(i) = \sum_{i=1}^{N}\left(\frac{\mu}{N} + \sigma\xi_N(i)\right).$$

Simplifying:

$$\ln\left(\frac{S_t}{S_0}\right) = \mu + \sigma W_N.$$

Thus:

$$\frac{S_t}{S_0} \approx e^{\mu+\sigma W_N}.$$

但是由于 μ 不依赖于 N，所以实际上等价于研究：

$$\lim_{N\to\infty} W_N$$

- 因为 W_N 是 N 个 i.i.d.，均值为 0、方差为 $\frac{1}{N}$ 的随机变量之和;
- 这个正好符合中心极限定理（CLT）的前提条件。

Distribution of W_N

The random variable W_N is the sum of N i.i.d. random draws of:

$$\begin{cases} \frac{1}{\sqrt{N}} & \text{with probability } \frac{1}{2}, \\ -\frac{1}{\sqrt{N}} & \text{with probability } \frac{1}{2}. \end{cases}$$

Limit Distribution

We are ultimately interested in the limiting behavior as $N \to \infty$:

$$\lim_{N\to\infty} e^{\mu+\sigma W_N}.$$

Since the exponential is continuous, this reduces to analyzing:

$$\lim_{N\to\infty} W_N.$$

By the Central Limit Theorem, W_N converges in distribution to a normal random variable. This limit is called:

$$W_t \quad \text{(standard Brownian motion)}.$$

Thus, in the continuous-time limit, we obtain the key building block for Brownian motion-based models like the Geometric Brownian Motion used in the Black-Scholes model.